

1 PHYSICAL DESIGN OF IoT (12 Marks)

Definition

Physical Design of IoT refers to the **actual hardware components and real-world objects** involved in an IoT system. It explains how devices are built, connected, and interact with the environment.

It focuses on:

- Sensors
 - Actuators
 - Embedded systems
 - Communication modules
 - Power supply
 - Connectivity interfaces
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Components of Physical Design of IoT

1. Devices (Things)

- Physical objects connected to the internet.
 - Have unique identity (IP address).
 - Perform sensing and actuation.
 - Example: Smart bulb, Smart watch.
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2. Sensors

- Detect physical/environmental parameters.

- Convert physical signals → digital data.
 - Examples:
 - Temperature sensor
 - Humidity sensor
 - Motion sensor
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3. Actuators

- Convert electrical signals → physical action.
 - Examples:
 - Motor
 - LED
 - Relay switch
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4. Embedded Systems / Microcontrollers

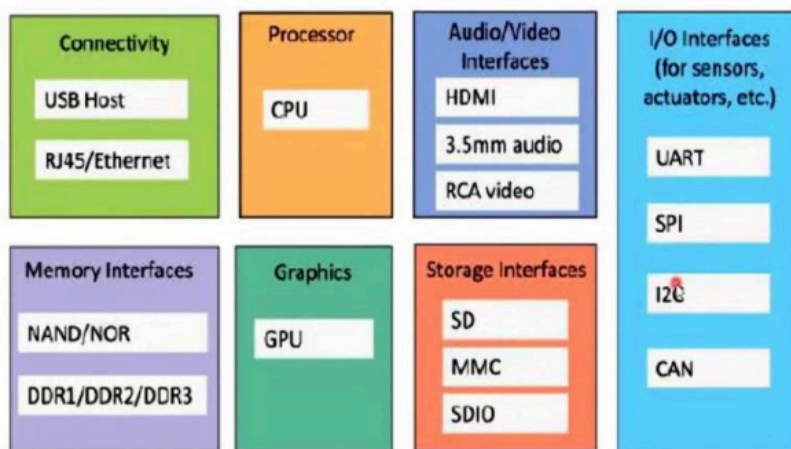
- Small computing units.
 - Process sensor data.
 - Control actuators.
 - Examples:
 - Arduino
 - Raspberry Pi
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5. Communication Modules

- Enable device connectivity.
 - Examples:
 - Wi-Fi
 - Bluetooth
 - ZigBee
 - Cellular (4G/5G)
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6. Power Supply

- Battery
- Solar panel
- Direct electric supply



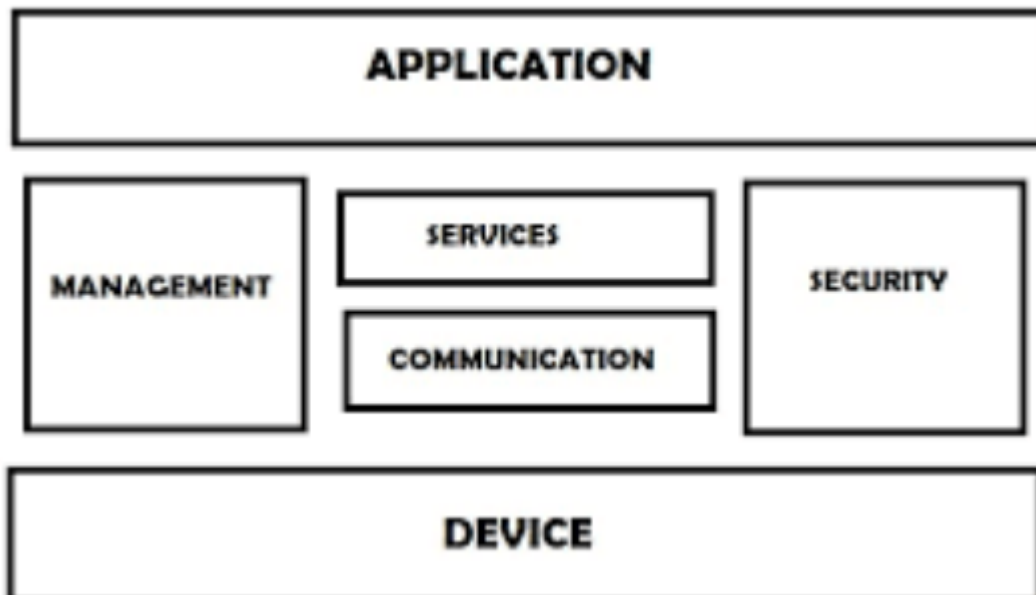
✓2 LOGICAL DESIGN OF IoT (Detailed – 12/15 Marks)

1. Introduction

Logical Design of IoT represents the **abstract functional architecture** of IoT systems. It focuses on:

- Data flow
- Functional blocks
- Communication models
- APIs
- Software architecture

Unlike physical design, it does NOT focus on hardware.



2. IoT Functional Blocks

2.1 Device

- Performs sensing and actuation.

2.2 Communication

- Transfers data between device and cloud.
- Uses protocols like MQTT, HTTP.

2.3 Services

Provides:

- Device monitoring
- Device discovery
- Data publishing
- Control services

2.4 Management

Handles:

- Configuration
- Device updates
- System control
- Fault management

2.5 Security

Ensures:

- Authentication
- Authorization

- Encryption
- Data integrity

2.6 Application

- User interface
- Dashboard
- Mobile app
- Web interface

3. IoT Communication Models

3.1 Request–Response Model

- Client sends request.
- Server sends response.
- Used in HTTP.

Example: Mobile app requesting temperature data.

3.2 Publish–Subscribe Model

Components:

- Publisher
- Broker
- Subscriber

Publisher → Broker → Subscribers

Used in MQTT.

Advantages:

- ✓ Decoupled communication

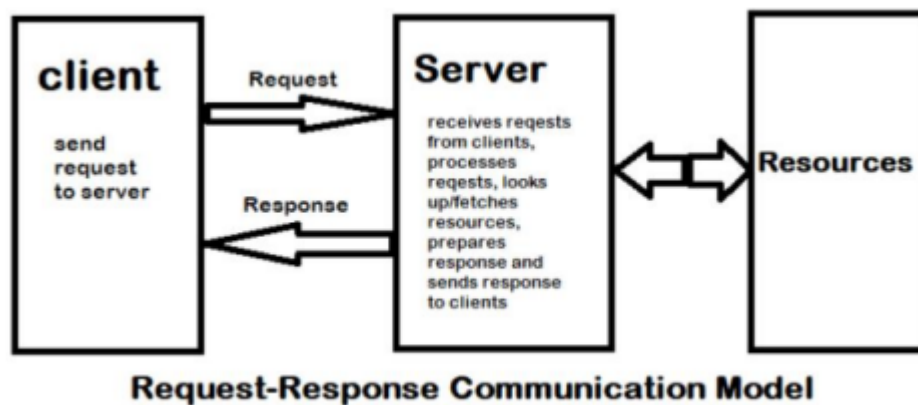
- ✓ Scalable
 - ✓ Efficient
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3.3 Push–Pull Model

- Producers push data to queue.
 - Consumers pull data from queue.
 - Used in distributed systems.
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3.4 Exclusive Pair Model

- Persistent connection.
- Full duplex communication.
- Used in WebSocket.
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4. IoT Communication APIs

REST API

- Uses HTTP methods

- Stateless communication
- Resource-based

WebSocket API

- Real-time communication
- Bi-directional
- Persistent connection

IoT LEVELS (Detailed – 12/15 Marks)

IoT Levels describe deployment architecture based on complexity.

Level 1

- Single node
 - Local storage
 - No cloud
 - Example: Smart home controller
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Level 2

- Single node
- Local analysis
- Cloud storage
- Example: Smart irrigation

Level 3

- Single node
- Cloud storage & analysis
- Example: Vibration monitoring

Level 4

- Multiple nodes
- Local analysis
- Cloud aggregation
- Example: Noise monitoring

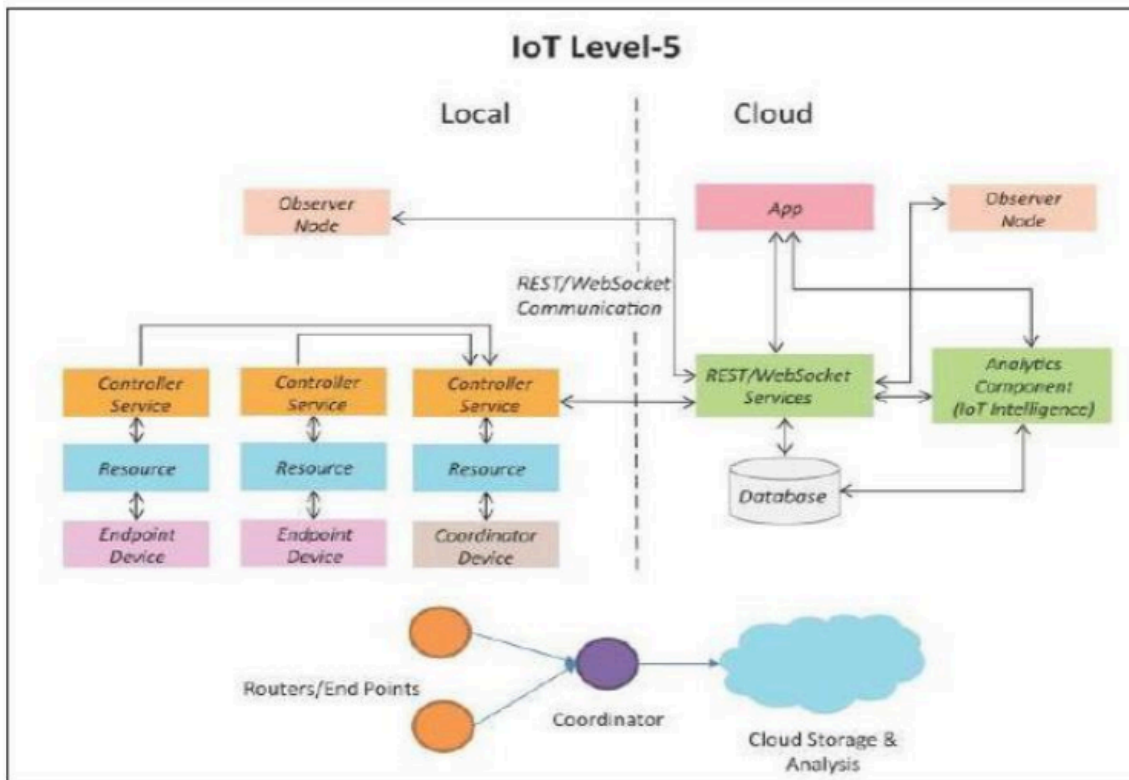
Level 5

- Multiple nodes
- One coordinator
- WSN-based
- Cloud storage
- Example: Forest fire detection

Level 6

- Multiple independent nodes

- Cloud analytics
- Centralized control
- Example: Weather monitoring system



UNIT - 2

✓1 IoT Core Technologies (Detailed Answer – 12/15 Marks)

1. Introduction

IoT Core Technologies are the fundamental communication protocols, standards, and connectivity solutions that enable IoT devices to connect, communicate, and exchange data reliably.

They form the backbone of IoT systems, ensuring:

- Reliable communication
- Real-time data transmission
- Secure connectivity
- Scalability

2. Core Networking Technologies in IoT

2.1 Wi-Fi (Wireless Fidelity)

- Common wireless LAN technology.
- High data rate.
- Medium power consumption.

Use in IoT:

- Smart home devices
- CCTV cameras
- Smart TVs

Example: Smart thermostat connected to home router.

2.2 Bluetooth & BLE (Bluetooth Low Energy)

- Short-range wireless technology.
- Very low power consumption.
- Suitable for battery-operated devices.

Use in IoT:

- Wearables
- Health monitoring systems

Example: Fitness tracker syncing with mobile phone.

2.3 Zigbee

- Low-power, low-data-rate wireless protocol.
- Supports mesh networking.
- Based on IEEE 802.15.4.

Use in IoT:

- Smart lighting
- Industrial sensors

Example: Smart lighting system in buildings.

2.4 LoRa (Long Range)

- LPWAN technology.
- Long-range communication (up to 15 km).
- Low power usage.

Use in IoT:

- Agriculture
- Environmental monitoring

Example: Soil moisture sensors in farms.

2.5 NB-IoT (Narrowband IoT)

- Cellular-based LPWAN.
- Wide coverage.
- Low energy consumption.

Use in IoT:

- Smart metering
 - Asset tracking
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2.6 Cellular Networks (3G/4G/5G)

- High-speed communication.
- Supports real-time applications.

Use in IoT:

- Smart cities
 - Connected vehicles
 - Industrial automation
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2.7 Ethernet & PoE

- Wired high-speed communication.
 - Very reliable.
 - Used in industrial IoT.
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2.8 MQTT (Message Queuing Telemetry Transport)

- Lightweight messaging protocol.
- Publish–Subscribe model.
- Ideal for low-bandwidth networks.

Use: Sensor-to-cloud communication.

2.9 CoAP (Constrained Application Protocol)

- Lightweight web protocol.
 - Designed for constrained devices.
 - Uses UDP.
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2.10 Edge & Fog Computing

- Data processing near source.
- Reduces latency.
- Reduces bandwidth usage.

Example: Autonomous vehicles making real-time decisions.

3. Applications of IoT Core Technologies

- Smart Homes
- Industrial IoT (IIoT)
- Smart Cities
- Agriculture Monitoring
- Healthcare IoT

3 IoT Supported Hardware (Detailed Answer – 12/15 Marks)

1. Introduction

IoT Supported Hardware platforms provide:

- Computing power
- Connectivity
- Interface support
- Storage

They are used for prototyping and deployment of IoT systems.

2. Raspberry Pi

- Single-board computer.
- Supports Linux OS.
- Has GPIO pins.

Features:

- Wi-Fi
- Bluetooth
- Ethernet
- USB
- HDMI

Applications:

- Smart home automation
 - Robotics
 - Surveillance systems
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3. ARM Cortex Processors

- Low-power microcontrollers.
- Energy efficient.
- Used in embedded IoT devices.

Types:

- Cortex-M (Microcontroller)
- Cortex-A (Application processor)

Applications:

- Wearables
 - Industrial sensors
 - Automotive systems
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4. Arduino Boards

- Open-source platform.
- Uses ATmega microcontrollers.
- Supports analog and digital I/O.

Features:

- Arduino IDE programming
- Low power
- Ideal for beginners

Applications:

- Smart lights
 - Environmental monitoring
 - Automatic irrigation systems
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5. Intel Galileo

- x86 architecture.
- Arduino compatible.
- More processing power.

Features:

- Linux support
- Ethernet
- USB
- Higher memory

Applications:

- Industrial IoT
 - Robotics
 - Data-intensive IoT systems
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6. Other IoT Hardware

- ESP8266 / ESP32 (Wi-Fi enabled)
- BeagleBone
- NVIDIA Jetson Nano (AI IoT)